Confirmation of the gravitational interaction in the business firms' network

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Abstract

We analyze Japanese inter-firm data containing about 700,000 firms' financial data and 4,450,000 transaction relation data between pairs of firms. We confirm that the amount of transaction is approximated by the product of fractional power of supplier's and buyer's annual sales, and we construct a generalized gravitational interaction model. All parameters of the model can be determined by data assimilation based on minimization of errors. The model has a wide applicability and we show some results of stress tests.

Keyword: Complex network, Gravitational interaction model

By analyzing the inter-firm business relation data it is reported that the network structure is characterized by a power law link number distribution which is explained by combination of two effects, preferential attachment and coagulation of firms [1]. As a next step of study on business network, we focus our attention to the flow of money.

The amount of money flow between business firms became observable recently owing to the data provided by Teikoku Databank, and it is known to be approximated by the product of fractional power of supplier's and buyer's annual sales [2] as typically shown in Fig.1.

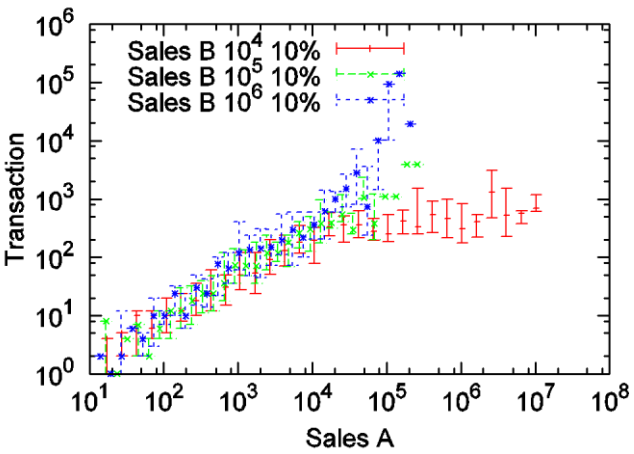


Fig.1 The amount of transaction money flow from firm A to firm B as a function of A's annual sales.

In this study, we estimate the parameters of the gravitational interaction model with an approach different from the previous work, a more straight forward method is provided. We compare the values of annual sales of firms estimated by this model and those of real sales, and confirm the validity of the model. Finally, as an application we conduct stress test on the assumption of various economic conditions with estimated parameter.

References

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